SCAMSHIELD: MACHINE LEARNING ARCHITECTURE & VIVA GUIDE

Comprehensive AI/ML Theory, Mathematical Foundations, Feature Engineering & Viva Defense
Project Context: Pune, Maharashtra, India | Continuous Technical Evaluation

30-SECOND ELEVATOR SUMMARY (THE OPENING DEFENSE STATEMENT):
*"ScamShield is a privacy-first, multilingual fraud detection system designed to protect vulnerable users from cyber scams across SMS, Email, and URLs. Instead of relying on a single slow black-box model, ScamShield deploys a **Modular Hybrid AI Architecture**: three specialized machine learning models (Word+Character n-gram Logistic Regression for SMS, Subject-Boosted Logistic Regression for Email, and a 22-feature Random Forest for URLs) calibrated alongside a deterministic Heuristic Rule Engine. The system achieves over **99% accuracy**, executes inference in under **15 milliseconds on CPU**, and provides actionable safety advice in English, Marathi, and Hindi."*

1. Problem Formulation & Architecture: Why a Modular Hybrid System?

ScamShield formulates fraud detection as **Supervised Binary Classification**: for any given input x, the system computes a continuous calibrated probability $P(y=1|x) \in [0, 1]$ that x is malicious. Rather than forcing a single model across heterogeneous vectors, ScamShield decouples the problem into three specialized pipelines.

Attack Vector	Syntactic Structure	Where Threat Lives	Why a Single Generic Model Fails
SMS / WhatsApp	Short (<160 chars), unpunctuated, Hinglish/Marathi slang	Urgent psychological coercion (light bill, lottery)	Generic NLP models treat regional slang & typos as out-of-vocabulary noise.
Email Phishing	Long-form (500-5000 chars), structured headers & body	High density in Subject line & call-to-action link	Long benign body text dilutes the few critical urgent threat keywords.
URL / Links	Purely syntactic string (RFC 3986), non-natural language	Shannon entropy, domain depth, high-risk TLDs	Language tokenizers fail on URLs or split them into meaningless byte fragments.

2. Algorithmic Justification: Classical Machine Learning vs. Deep Learning / LLMs

Evaluation Criterion	ScamShield Classical ML (Current)	Deep Learning / Large Language Models (BERT/LLM)	Engineering Verdict
Inference Latency	<15 milliseconds on basic CPU	500 - 3,000 milliseconds (requires GPU)	**PASS: Essential for real-time mobile/shopkeeper UPI checks**
Memory & Binary Size	<50 MB RAM; ~3 MB model binaries (.pkl)	2 GB - 10 GB VRAM required	**PASS: Runs on low-cost cloud tiers and edge devices**
Mathematical Explainability	Direct log-odds weights & Gini feature importance	Opaque multi-billion parameter black-box	**PASS: Enables plain-language trilingual XAI explanations**
Adversarial Zero-Day Defense	Compounded with deterministic Heuristic Rule Engine	Vulnerable to prompt injection & jailbreaking	**PASS: Deterministic rules guarantee zero-tolerance security**

3. Mathematical Foundations of the Deployed Algorithms

A. LOGISTIC REGRESSION WITH L2 REGULARIZATION (SMS & EMAIL):
Logistic Regression maps linear combinations of input features into calibrated probabilities [0, 1] using the **Sigmoid (Logistic) Function**:

- Linear combination: $z = w^T x + b = \sum w_j x_j + b$
- Probability output: $P(y = 1 \mid x) = \sigma(z) = 1 / (1 + e^{-z})$
- **Penalized Log-Loss (Binary Cross-Entropy) Objective:**
  $L(w) = - (1/N) \sum [y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)] + (1 / 2C) \|w\|_2^2$
- *Why this works:* Convex loss surface guarantees global convergence. High-dimensional sparse TF-IDF spaces (14,000 dimensions) have linear decision boundaries that prevent overfitting. The inverse regularization parameter C was tuned to **C=5.0** for SMS and **C=3.0** for Email.

B. RANDOM FOREST CLASSIFIER WITH GINI IMPURITY (URL LINKS):
Random Forest is an ensemble of $B = 100$ decision trees trained on bootstrap samples with random feature subsets:

- Ensemble Probability: $\hat{P}(y = 1 \mid x) = (1 / B) \sum T_b(x)$
- **Gini Impurity Splitting Criterion:** $I_G(p) = 1 - (p_{safe}^2 + p_{scam}^2)$ (pure leaf = 0.0, split = 0.5)
- *Hyperparameter constraints:* **max_depth=15** and **min_samples_split=5** prevent memorization of training URLs.

4. Feature Engineering Innovations: Dual-Channel Subwords & 22 URL Features

SMS Subword FeatureUnion: Standard word tokenizers fail on intentional typos (*p@ypal, s-b-i*) and Hinglish/Marathi transliteration (*paise, light bill*). ScamShield uses **FeatureUnion** combining Word-level TF-IDF (1-3 ngrams, 7,000 features) and Character-boundary TF-IDF (char_wb, 3-5 ngrams, 7,000 features). Subword chunks match the root scam pattern regardless of spelling variations, **boosting Recall from 88.59% to 94.63%**.
Email Subject-Line Boosting: In phishing, the subject line contains 2-3x higher scam intent density than the body. ScamShield double-weights the subject (*f"{subject} {subject} {body}"*), doubling subject feature frequencies without distorting body context.

Complete 22-Feature URL Structural Space (Zero Network Requests):

Feature Group	Feature Name	Type	Threat Detection Rationale
Lengths	url_length, hostname_length, path_length, query_length	Integer	Malicious URLs elongate paths to conceal domains or pass encoded attack payloads.
Delimiters	count_dots, count_hyphens, count_at, count_slash, count_equal	Integer	Chained hyphens (sbi-yono-update) and '@' symbols indicate credential injection.
Structural	subdomain_count, count_digits, count_special_chars	Integer	Subdomain stacking (paypal.com.verify-login.xyz) mimics official hosts.
Security Flags	has_https, has_ip_address, is_shortened, suspicious_keyword_count	Binary/Int	Flags raw IPv4 links, URL shorteners (bit.ly), and banking keywords (kyc, otp, login).
Advanced Domain	entropy (Shannon Entropy of Hostname)	Float	H(S) = -∑ p(c) log2 p(c). High entropy (>3.8) detects auto-generated DGA phishing domains.
Advanced Domain	has_high_risk_tld	Binary	Flags heavily abused free/cheap extensions (.xyz, .top, .buzz, .cfd, .rest, .tk, .icu).
Targeted Brand	brand_keyword_count, host_hyphen_count, digit_ratio_host	Int/Float	Catches targeted Indian brand lures (sbi, yono, paytm, mahadiscom) with numeric obfuscation.

5. The Hybrid Risk Calibration Engine: Blending ML with Deterministic Rules

Machine learning models provide continuous probabilities based on statistical patterns. However, cybersecurity requires strict zero-tolerance safeguards. ScamShield pairs the ML models with a **Heuristic Rule Engine** through late-stage linear fusion:

HYBRID RISK SCORING FORMULAS:
- **With Embedded URL:** $Risk\ Score = 0.45 \times (P_{Text} \times 100) + 0.35 \times (P_{URL} \times 100) + 0.20 \times S_{Rule}$
- **Text Only (SMS / Email):** $Risk\ Score = 0.60 \times (P_{Text} \times 100) + 0.40 \times S_{Rule}$
- **Standalone URL:** $Risk\ Score = 0.70 \times (P_{URL} \times 100) + 0.30 \times S_{Rule}$
  - P_{Text}, P_{URL} *in [0, 1]*: Continuous model probabilities.
  - S_{Rule} *in [0, 100]*: Cumulative penalty for deterministic red flags (Raw IP +40, Sender mismatch +35, Shortener +25, Power cut threat +50).

Calibrated Risk Tiers: Low Risk (0 - 39): Safe / verified | **Medium Risk (40 - 69)**: Cautionary warning | **High Risk (70 - 100)**: Active scam

6. Empirical Evaluation Metrics & Confusion Matrices

All models were evaluated on independent, held-out test distributions (20% stratified test split). In fraud detection, **Recall is the primary safety metric** because a False Negative (missed scam) leads to financial catastrophe, whereas a False Positive merely triggers a cautionary alert.

Module	Algorithm & Configuration	Test Size	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SMS Message	FeatureUnion TF-IDF + Logistic Reg (C=5.0)	1,115	**99.19%**	99.30%	**94.63%**	**96.91%**	0.9944
Email Phishing	Subject-Boosted TF-IDF + Logistic Reg (C=3.0)	8,000	**98.22%**	97.89%	**98.58%**	**98.23%**	0.9979
URL / Links	Random Forest (100 trees, depth 15, 22 feats)	8,000	**99.05%**	**99.90%**	**98.20%**	**99.04%**	0.9984

Held-out Test Confusion Matrices:

Model	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)	Clinical Interpretation
SMS Classifier (N=1,115)	965 (Safe)	1 (Safe flagged)	8 (Scam missed)	**141 (Scam caught)**	Only 1 false alarm out of 966 safe texts; caught 94.6% of scams.
Email Classifier (N=8,000)	3,915 (Safe)	85 (Safe flagged)	57 (Phish missed)	**3,943 (Phish caught)**	Caught 3,943 out of 4,000 phishing attacks; precision 97.9%.
URL Classifier (N=8,000)	3,996 (Benign)	4 (Benign flagged)	72 (Malicious missed)	**3,928 (Malicious caught)**	Exceptional 99.9% precision; only 4 false alarms across 4,000 benign URLs.

7. Viva Defense & Technical Evaluation: Top 12 Questions & Answers

Carefully study these 12 questions; they represent the exact technical topics examiners interrogate during project defense:

Q1: Why is Recall prioritized over Precision in cyber scam detection?

• *Model Answer:* In fraud detection, a False Negative means a victim receives an active scam, believes it, and loses money. A False Positive merely causes minor friction by showing a cautionary warning on a safe message. High Recall (>94% on SMS, >98% on Email) ensures virtually no real scams slip through.

Q2: How did you prevent data leakage during training?

• *Model Answer:* We performed stratified 80/20 train/test splitting before applying any feature extraction or scaling. Vectorizers (TfidfVectorizer, FeatureUnion) and scalers were fitted strictly on X_train using fit_transform(), and then applied to X_test using transform() alone. No test data informed the vocabulary or parameters.

Q3: If an attacker uses a valid HTTPS link, how does your URL model catch it?

• *Model Answer:* Over 80% of modern phishing sites use free SSL certificates (Let's Encrypt). Our model does not rely solely on has_https; it incorporates Shannon entropy (detecting random auto-generated domains), High-risk TLD tracking (.xyz, .top), Indian brand lure scanning (sbi, yono, paytm in host), and subdomain stacking count.

Q4: How does the SMS model handle Hinglish and Marathi code-mixing?

• *Model Answer:* We implemented a FeatureUnion combining word n-grams (1-3) with character-boundary n-grams (char_wb, 3-5 characters). Subword fragments match the morphological patterns of regional transliterations (paise, bhejo, light bill) and typo variations, even if the complete word is not in an English dictionary.

Q5: Why did you use TF-IDF with sublinear_tf=True instead of simple CountVectorizer?

• *Model Answer:* CountVectorizer only counts raw frequency, allowing common uninformative words to dominate. TF-IDF boosts words that are unique to scam documents (suspended, kyc, lottery). Setting sublinear_tf=True (1 + ln(tf)) prevents long messages from dominating feature weights simply by repeating threat words.

Q6: What does the hyperparameter C do in Logistic Regression?

• *Model Answer:* C is the inverse of regularization strength (C = 1/lambda). A small C applies heavy shrinkage to weights to prevent overfitting; a large C fits training features more closely. We tuned C=5.0 for SMS and C=3.0 for Email to maximize recall while maintaining low false-positive rates.

7. Viva Defense & Technical Evaluation (Continued)

Q7: What is Shannon Entropy and how does it detect malicious URLs?

• *Model Answer:* Shannon entropy measures information uncertainty and character randomness: H(S) = -∑ p(c) log2 p(c). Legitimate domains (google.com, apple.com) have low entropy (1.5 - 2.8) due to phonetic redundancy. Auto-generated DGA domains (x7k9-p4zm1.com) have high entropy (>3.8).

Q8: Why did you restrict max_depth=15 in Random Forest?

• *Model Answer:* Without max_depth, decision trees expand until all leaves are completely pure, causing extreme overfitting, bloated model sizes (>50 MB), and slow inference. Restricting to depth 15 with min_samples_split=5 ensures generalizability, keeps inference <5ms, and limits binary size to ~3 MB.

Q9: How do you handle class imbalance in the SMS dataset?

• *Model Answer:* In the raw SMS dataset, only 13.4% are spam and 86.6% are ham. A naive model would predict 'Safe' 100% of the time and achieve 86.6% accuracy while catching zero scams. We resolved this using class_weight='balanced', which multiplies the loss penalty for minority scam samples inversely proportional to class frequencies.

Q10: Why do you avoid making live network requests during URL analysis?

• *Model Answer:* 1. User Safety: Visiting malicious links triggers drive-by malware or confirms active victim addresses to scammers. 2. Latency: Network round-trips take 1000 - 3000ms; local lexical extraction takes <2ms. 3. Privacy: The user's browsing activity is never shared with third parties.

Q11: What is the purpose of the OCR vision module and how does it integrate with ML?

• *Model Answer:* Elderly users and shopkeepers often receive fraud via WhatsApp screenshots rather than text. We use Tesseract OCR with OpenCV adaptive thresholding and grayscale conversion. The extracted text is automatically classified as SMS or URL and routed to the corresponding ML predictor seamlessly.

Q12: How is client privacy protected before machine learning inference?

• *Model Answer:* ScamShield includes a client-side Privacy Redaction Engine that scrubs Indian PII (Aadhaar 12-digit numbers, PAN 10-digit IDs, UPI VPA handles, and credit card numbers) using regular expressions before saving logs or sending text to inference routines.

8. Codebase Directory & File Inventory for Technical Review

Component	File Path	Key Class / Function
SMS Model Training	ml/train_sms_model.py	train_sms_model() with FeatureUnion
Email Model Training	ml/train_email_model.py	train_email_model() with Subject Boosting
URL Model Training	ml/train_url_model.py	train_url_model() with 22 features
Feature Engineering	ml/feature_extraction.py	extract_url_features(), shannon_entropy()
Inference Predictors	ml/prediction.py	SMSPredictor, EmailPredictor, URLPredictor
Hybrid Risk Engine	risk/risk_engine.py	compute_hybrid_risk() with calibrated weights
Privacy Redaction	risk/privacy.py	mask_sensitive_data() (Aadhaar, PAN, UPI)
Evaluation Runner	ml/evaluate_models.py	evaluate_all() generating overall summary metrics